

App Development Project Report

1. Student Details

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2. Project Details

Project Title: Placement Portal Application - V2

Problem Statement:

To design and build a web-based application that allows **Admin (Institute), Companies, and Students** to interact with the application based on their roles, such as managing placement drives, tracking applications and recruitments.

Approach:

The app was built using VueJS and ChartJS in the frontend, and Flask, Redis and Celery in the backend. It allows users to log placement activities, visualize trends through summary charts, and analyze company insights over time.

3. AI/LLM Declaration

I used **GitHub Copilot and ChatGPT** to assist in writing SQLAlchemy model definitions.

The extent of AI/LLM usage is around **25–30%**, limited to **code and error fixing suggestions**. All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
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Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Chart.js	To visualize trends such as applied vs selected
VueJS	UI
JWT	User authentication
Redis	Caching
Redis and Celery	Batch Jobs
SQLite	Lightweight local database for storing user data

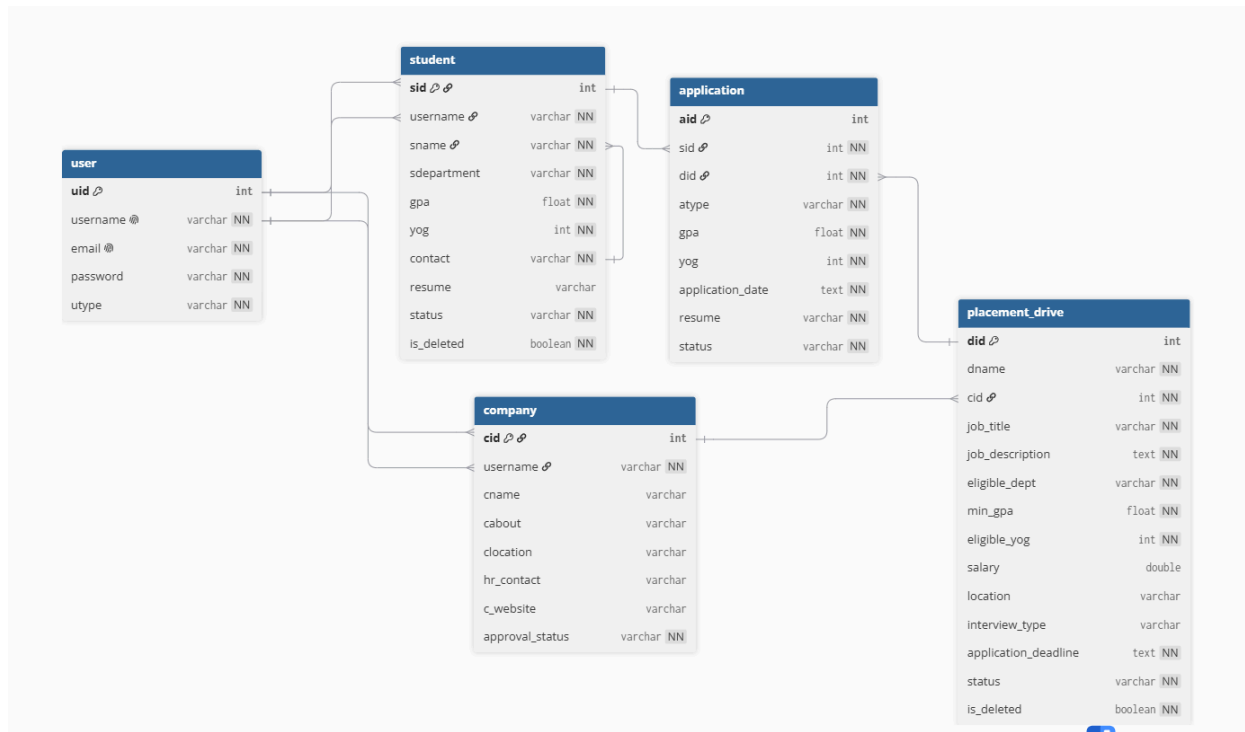
5. Database Schema / ER Diagram

Relations:

1. **User** — stores user profile details (uid, username, email, password, type)
2. **Student** — logs student profile details (sid, username, sname, sdepartment, gpa, yog, contact, resume, status)
3. **Company** — logs company profile details (cid, username, cname, cabout, clocation, hr_contact, c_website, approval_status)
4. **Placement_Drive** — stores drive details (did, dname, cid, job_title, job_description, eligible_dept, min_gpa, eligible_yog, salary, location, interview_type, application_deadline, status, is_deleted)
5. **Application** — stores student application details (aid, sid, did, type, gpa, yog, application_date, resume, status)

Relationships:

- One-to-One → User → Student
- One-to-One → User → Company
- One-to-Many → Company → Placement_Drive
- One-to-Many → Placement_Drive → Application
- One-to-Many → Student → Application



6. API Resource Endpoints

Endpoint	Method	Description
/api/admin/login	POST	Authenticate admin and generate session
/api/student/login	POST	Authenticate student and generate session
/api/company/login	POST	Authenticate company and generate session
/api/export/applications	GET	Get the placement history of student

YAML API Definition File:

Included separately in the submission ZIP as [api.yaml](#).

7. Architecture and Features

Architecture Overview:

- **/frontend** – Vue components, views, router, assets
- **/backend** – Flask api routes

Implemented Features:

- Admin is the pre-existing superuser
 - Registration for company and students, and login for all (admin, company, student)
 - Student applications maintenance and placement history logging
 - No multiple applications by the same student for the same drive
 - Visualization dashboard using [Chart.js](#)
 - Secure password hashing and user session handling
 - Summary analytics for admin, company and student
 - Monthly Placement Report for admin as scheduled job
 - Daily Reminders for student as scheduled job
 - User Triggered Async Job – Export Applications for student
 - Caching is implemented for performance
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8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1tLYft4i2RfJ7S88Cp4STcnTI_DqO5rf0/view?usp=sharing
